

Why the Poor Do Not Expropriate the Rich (1998)

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19.05.2026

Motivation

- Classical fear of political elites:
 - once poor citizens receive voting rights,
 - they will vote to expropriate the rich.
- Basic intuition:
 - median wealth is below average wealth,
 - therefore the majority should support expropriation
- However, universal suffrage in the twentieth century did not lead to full redistribution.

Key Idea

Voters may vote against their economic interests if they strongly care about cultural issues.

Assumption (Redistributive tax economy)

$$W = [\underline{w}, \bar{w}], \quad \text{average wealth} = m = \int_{\underline{w}}^{\bar{w}} wg(w) dw, \quad t \in [0, 1].$$

A voter with wealth w has indirect utility

$$v(t; w) = (1 - t)w + tm = w + t(m - w).$$

$$\frac{\partial v(t; w)}{\partial t} = m - w \implies \begin{cases} w < m & \Rightarrow \text{strictly prefers higher } t, \\ w > m & \Rightarrow \text{strictly prefers lower } t. \end{cases}$$

One-Dimensional Primitives

Assumption (Electoral system)

$$\Pi^1(\tau^1, \tau^2) = p_L(\tau^1, \tau^2)v(\tau^1; w_L^1, a_L) + (1 - p_L(\tau^1, \tau^2))v(\tau^2; w_L^1, a_L)$$

$$\Pi^2(\tau^1, \tau^2) = p_L(\tau^1, \tau^2)v(\tau^1; w_R^1, a_R) + (1 - p_L(\tau^1, \tau^2))v(\tau^2; w_R^1, a_R).$$

Assumption (Parties)

Left represents $w_L < m$ and Right represents $w_R > m$.

Assumption (Voters)

Distribution of voters is $g_s(w)$, where s is a random variable (state) uniformly distributed on $[0, 1]$. The mean of g_s is μ . The C.D.F. of g_s by G_s and G_s is strictly decreasing in s

Voting Probability in One Dimension

When you win?

Given s , $G_s(\mu)$ is the fraction of voters who vote for t^1 over t^2 , where $t_1 > t_2$, in state s . Now t^1 defeats t^2 just in case this is a majority, i.e., when

$$G_s(\mu) > \frac{1}{2}.$$

Assumption (Winning Possibility)

There exists such s^ that $G_{s^*}(\mu) = \frac{1}{2}$*

Thus Left's win probability is

$$p_L(t_L, t_R) = \begin{cases} s^*, & t_L > t_R, \\ 1 - s^*, & t_L < t_R, \\ \frac{1}{2}, & t_L = t_R. \end{cases}$$

Partisan Payoffs

Expected party payoffs are

$$\Pi_L(t_L, t_R) = p_L(t_L, t_R)v(t_L; w_L) + (1 - p_L(t_L, t_R))v(t_R; w_L),$$

$$\Pi_R(t_L, t_R) = p_L(t_L, t_R)v(t_L; w_R) + (1 - p_L(t_L, t_R))v(t_R; w_R).$$

Since

$$v(t; w_L) = w_L + t(m - w_L) \text{ is increasing in } t,$$

and

$$v(t; w_R) = w_R - t(w_R - m) \text{ is decreasing in } t,$$

the two parties have opposite ideal tax rates:

$$t_L^{\text{ideal}} = 1, \quad t_R^{\text{ideal}} = 0.$$

Stackelberg: Right Moves, Left Responds

If $t_R < 1$, then for any $t_L > t_R$,

$$\Pi_L(t_L, t_R) = s^* v(t_L; w_L) + (1 - s^*) v(t_R; w_L).$$

Since $m - w_L > 0$,

$$\frac{\partial \Pi_L}{\partial t_L} = s^*(m - w_L) > 0 \quad \text{on } (t_R, 1].$$

Thus

$$\text{BR}_L(t_R) = 1 \quad (t_R < 1).$$

If $t_R = 1$, no feasible tax exceeds t_R . Is $t_L = 1$ still the best response?

$$m \geq (1 - s)(w + t(m - w)) + sm \tag{1}$$

$$m(1 - s)(1 - t) \geq (1 - s)(1 - t)w \tag{2}$$

$$m \geq w \tag{3}$$

so again

$$\text{BR}_L(1) = 1.$$

$$\text{BR}_L(t_R) = 1 \quad \forall t_R \in [0, 1].$$

Stackelberg: Left Moves, Right Responds

If $t_R < t_L$, Right wins with probability $1 - s^*$, so

$$\Pi_R(t_L, t_R) = s^* v(t_L; w_R) + (1 - s^*) v(t_R; w_R).$$

Since $m - w_R < 0$,

$$\frac{\partial \Pi_R}{\partial t_R} = (1 - s^*)(m - w_R) < 0.$$

Hence Right's best response in the lower-tax region is $t_R = 0$.

In case of choice, $t_R > t_L$ there exist the profitable deviation in the form of $t_R = 0$. The condition for deviation always hold

$$s^* \leq \frac{t_L}{2t_L - t_R}$$

Therefore

$$\text{BR}_R(t_L) = 0 \quad \forall t_L \in [0, 1].$$

Anticipating $t_R = 0$, Left solves

$$\max_{t_L \in [0, 1]} \{s^* v(t_L; w_L) + (1 - s^*) v(0; w_L)\},$$

and the objective is increasing in t_L . Thus

$$t_L^* = 1, \quad t_R^* = 0.$$

One-Dimensional Proposition

Proposition

Under the one-dimensional assumptions, whether Left or Right is the Stackelberg follower,

$$t_L^* = 1.$$

If Right moves first:

$$BR_L(t_R) = 1 \quad \forall t_R.$$

Thus any Stackelberg equilibrium has

$$t_L^* = 1.$$

If Left moves first:

$$BR_R(t_L) = 0 \quad \forall t_L,$$

and Left maximizes

$$s^* [w_L + t_L(m - w_L)] + (1 - s^*)w_L.$$

Since

$$\frac{d}{dt_L} = s^*(m - w_L) > 0,$$

again $t_L^* = 1$.

Two-Dimensional Policy Space

The policy space is

$$\mathcal{T} = [0, 1] \times \mathbb{R}, \quad \tau_i = (t_i, z_i).$$

A voter with traits (w, θ) has utility

$$v(t, z; w, \theta) = (1 - t)w + tm - \frac{\alpha}{2}(z - \theta)^2,$$

where $\alpha > 0$ is the salience of the non-economic issue.

Define

$$\Delta t = t_R - t_L, \quad \Delta z = z_R - z_L, \quad \bar{z} = \frac{z_L + z_R}{2}.$$

For $\Delta z > 0$, a voter prefers Left to Right iff

$$\Delta t(w - m) + \alpha \Delta z (\bar{z} - \theta) > 0,$$

equivalently

$$\theta < \bar{z} + \frac{\Delta t(w - m)}{\alpha \Delta z}.$$

Electoral Probability in Two Dimensions

In state s , voter traits have density

$$h_s(w, \theta) = g_s(w)r(\theta | w).$$

For $\Delta z > 0$, the Left coalition has measure

$$H_s(W_L) = \int_W \int_{-\infty}^{\bar{z} + \frac{\Delta t(w-m)}{\alpha \Delta z}} g_s(w)r(\theta | w) d\theta dw.$$

The winning threshold $s^*(\tau_L, \tau_R)$ solves

$$H_{s^*}(W_L) = \frac{1}{2},$$

and therefore

$$p_L(\tau_L, \tau_R) = s^*(\tau_L, \tau_R).$$

As $\alpha \rightarrow \infty$,

$$\bar{z} + \frac{\Delta t(w-m)}{\alpha \Delta z} \rightarrow \bar{z},$$

so the election becomes asymptotically a contest over z .

Party Factions

Left's three factional objectives are

$$\begin{array}{ll} \text{militants:} & v_L(\tau_L), \\ \text{opportunists:} & p_L(\tau_L, \tau_R), \\ \text{reformists:} & \Pi_L(\tau_L, \tau_R) = p_L v_L(\tau_L) + (1 - p_L) v_L(\tau_R). \end{array}$$

Right's analogous objectives are

$$v_R(\tau_R), \quad 1 - p_L(\tau_L, \tau_R), \quad \Pi_R(\tau_L, \tau_R) = p_L v_R(\tau_L) + (1 - p_L) v_R(\tau_R).$$

Thus a platform change must satisfy simultaneous inequalities, not a single maximization problem.

PUNE as a Nash Equilibrium with Incomplete Preferences

Left:

$$(\tau'_L, \tau_R) K_L(\tau_L, \tau_R)$$

iff

$$\begin{cases} p_L(\tau'_L, \tau_R) \geq p_L(\tau_L, \tau_R), \\ v_L(\tau'_L) \geq v_L(\tau_L), \\ \Pi_L(\tau'_L, \tau_R) \geq \Pi_L(\tau_L, \tau_R). \end{cases}$$

Right:

$$(\tau_L, \tau'_R) K_R(\tau_L, \tau_R)$$

iff

$$\begin{cases} 1 - p_L(\tau_L, \tau'_R) \geq 1 - p_L(\tau_L, \tau_R), \\ v_R(\tau'_R) \geq v_R(\tau_R), \\ \Pi_R(\tau_L, \tau'_R) \geq \Pi_R(\tau_L, \tau_R). \end{cases}$$

$(\tau_L, \tau_R) \text{ is PUNE} \iff \nexists K_L\text{-improving deviation and } \nexists K_R\text{-improving deviation.}$

Limit Condition for the PUNE Result

Consider a non-trivial PUNE of the limiting $\alpha = \infty$ game:

$$(z_L^\infty, z_R^\infty), \quad 0 < p_L(z_L^\infty, z_R^\infty) < 1.$$

Let

$$\Delta z^\infty = z_R^\infty - z_L^\infty, \quad \bar{z}^\infty = \frac{z_L^\infty + z_R^\infty}{2},$$

and define the density weight on the decisive z -cohort by

$$q(w) = g_{s^*}(w)r(\bar{z}^\infty | w).$$

The mean wealth of that cohort is

$$\bar{m} = \frac{\int_W wq(w) dw}{\int_W q(w) dw}.$$

The key inequality, which if holds make left propose tax equal to 0, is:

$$\boxed{\bar{m} - m > \frac{(m - w_L)\Delta z^\infty}{2(z_L^\infty - \theta_L)}}. \quad (\text{A3}^*)$$

From A3* to the Main PUNE Theorem

If electoral uncertainty over the median z -voter is small, then for every non-trivial limiting PUNE,

$$\Delta z^\infty = z_R^\infty - z_L^\infty < \varepsilon.$$

Therefore the right-hand side of (A3*) is small:

$$\frac{(m - w_L)\Delta z^\infty}{2(z_L^\infty - \theta_L)} \approx 0.$$

If the decisive median- z cohort satisfies

$$\bar{m}_{\text{median } z} > m,$$

$\alpha \text{ large} \implies t_L = 0 \text{ in every non-trivial PUNE.}$
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U.S. Results

